

larisation filter to separate the left and right images.

"Both the dual projector and colour wheel technologies require silver screen or ZScreen—a high-definition (HD) screen that preserves the light and doesn't dissipate it," says Vardhan. "Regular screen cannot preserve the light polarisation that is used by both the technologies," adds Senthil. The screen is a mirrored surface, silver in colour, hence the name silver screen.

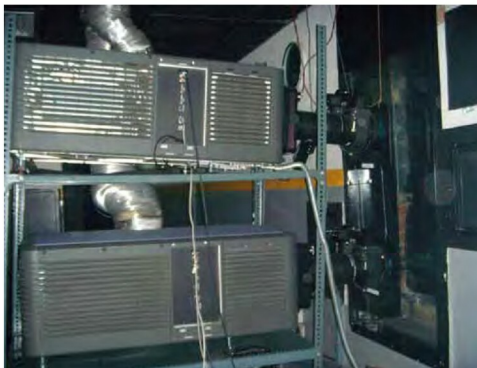
Single-projector-based XpanD technology is apparently the most easy for 2-D to 3-D transition. The benefit of this technology is that it does not require a silver screen. This greatly saves the cost involved in second projector or the silver screen. However, you need active filter glasses with electronics to view the image. There are infrared (IR) emitters in the auditoria that are synchronised with the active left and right eye-glasses to open and shut them (glasses) alternately. When a left-eye image is projected, the infrared emitter sends a signal to open the left-side glass and shut the right and vice versa.

In terms of quality and performance, all the technologies are almost equal. "We have experimented and found that with the same intensity of digital projector, the dual projector and colour wheel technologies are more suitable for big theatres and auditoria than the XpanD, which works well in small areas. So we have implemented XpanD in the PVR Gold," inform Rajiv and team from PVR.

Digital projectors

"Digital light processing (DLP) has a predominant market share in projection technology for 3-D cinema," says Ganesh. Its slight variant DLP cinema technology adheres to the Digital Cinema Initiatives (DCI) standards laid by a team of Hollywood studios. The DCI standards were introduced to ensure that the quality is not compromised while transitioning to digital.

The DLP technology from Texas Instruments can project 3-D images



Dual projector arrangement (courtesy: PVR Cinemas, Delhi)

using a single projector. TI's DLP chip is used in digital projectors. "Marco, NEC and Christie are the leading manufacturers of DLP cinema projectors," shares Ganesh.

Explaining the technology, Ganesh says, "DLP is an optical semiconductor that has millions of tiny mirror particles. It is called DLP or DMD (digital micro-mirror device). This optical semiconductor manipulates light using images that are sent out from the server and the image is formed on the screen." The tiny mirrors are individually controllable and move extremely fast. When the coloured light falls on this chip, the image is generated on the chip and magnified using optical components to project on the large screen.

Texas Instruments offers two chipset with different resolutions, 2K and 4K. "2K chip is used for screens normally up to 21.33m (70-feet) size and 4K does well for screens up to 30.48m (100-feet) wide," says Ganesh. "Around 14,000 cinema theatres have adopted DLP technology worldwide, out of which 7000 are 3-D-enabled.

Out of 14,000, approximately 200 are in India, and nearly 50 of them are 3-D-enabled."

Cinema playback servers

In digital cinema, you need a digital playback server to play the digital video file. "Digital cinema server is a computer, essentially a secured playback device," explains Senthil. The digital movie is loaded on the server for playback.

When you get the digital movie file in DCI-compliant J2K format, the server first loads the file on its internal storage. While playing, it decrypts the movie as the file is encrypted. "It puts the water mark (date and time) on the movie and then encrypts it back. The encrypted signal is sent to the projector over a serial link called high-definition serial data interface (HDS DI)," says Senthil. Unless the projector knows decryption keys, which are sent over Ethernet, it will not be able to decrypt the video. This protects against piracy.

QubeXP-D from Qube is one of the commonly used servers. It is based on